

17 Carbonyl compounds

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Aldehydes and ketones have polar C=O bonds susceptible to nucleophilic addition. Aldehydes oxidise with Tollens' or Fehling's reagents while ketones generally do not. NaBH₄ reduces aldehydes to primary alcohols and ketones to secondary alcohols. HCN addition creates cyanohydrins and may create a chiral centre.

Worked example: follow the reasoning

Propanal is reduced to propan-1-ol with NaBH₄; propanone gives propan-2-ol.

Quick check · multiple choice

A Which carbonyl compound forms a silver mirror with Tollens' reagent? [1]

- A Aldehyde
- B Ketone only
- C Alkane
- D Ester only

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 An unknown C₃H₆O compound forms a silver mirror and is reduced to propan-1-ol. [3]
Identify it and justify.

Answers and explanation

Quick check: A. Aldehyde

Question 1 [3 marks]

Propanal [1]; aldehyde reduces Tollens' reagent [1]; carbonyl reduction gives primary alcohol [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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